GOKUL D

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student (CGPA 8.38/10) with hands-on experience in Embedded C firmware on ESP32, STM32 and Arduino, including sensor interfacing, BLE and cloud-connected IoT systems. Completed two industrial placements in marine electronics and automation, including service support on navigation and communication equipment aboard INS Vikrant. Moving into Edge AI and computer vision, with a MobileNetV3 classifier exported to ONNX and served through Fast API for NVIDIA Jetson deployment. Seeking embedded, firmware, IoT or electronics R&D internships.

TECHNICAL SKILLS

Programming: Embedded C, C++, Python, Java, JavaScript, Dart
Microcontrollers & Embedded: ESP32, ESP32-CAM, STM32, Arduino; GPIO, PWM, UART, I²C, SPI, BLE, sensor interfacing, actuator control, firmware development
IoT & Connectivity: ESP-NOW, BLE, REST API, Firebase, JSON
Edge AI & Computer Vision: OpenCV, ONNX, AI model integration, LLM/API integration
Backend & Applications: Flask, Fast API, REST APIs, Flutter
Electronics: Circuit debugging, PCB-related work, control panel wiring, marine electronics
Tools & Platforms: Git, GitHub, VS Code, Arduino IDE, Android Studio; Linux (Ubuntu), Windows

EDUCATION

B.E. Electronics and Communication Engineering 2024 – 2028
Adithya Institute of Technology, Coimbatore (Anna University) CGPA: 8.38 / 10

INDUSTRIAL / INTERNSHIP EXPERIENCE

Assistant to Senior Service Engineer, Navicom Technology International Pvt. Ltd. *May 25, 2026 – June 05, 2026*
Marine Electronics, Navigation & Communication Systems | INS Vikrant

- Assisted a senior service engineer with servicing and maintenance of marine navigation and communication equipment aboard INS Vikrant.
- Supported troubleshooting of shipboard electronic systems under engineer supervision.
- Gained exposure to service procedures for defence-related electronics in an operational naval environment.

Industrial Trainee, SMEC Automation Pvt. Ltd. *June 05, 2025 – June 20, 2025*
Marine Automation

- Built control panels for marine automation: component mounting, electrical wiring and termination.
- Followed electrical schematics during panel assembly and tested completed panels.

SELECTED ENGINEERING PROJECTS

- **Sat Query AI:** Built an edge AI satellite-image classification pipeline using MobileNetV3-Small, achieving 97.78% validation accuracy and deploying the ONNX model through Fast API for NVIDIA Jetson.
- **AMPET:** Developed an ESP32-based smart energy monitoring and theft-detection system integrating voltage/current sensors with a Flask web dashboard.
- **VENOM**: Developed an AI-powered robotic assistant combining Arduino, PCA9685 servo control, OpenCV and AI-based interaction.
- **RAV – Reasoning-Aware Vision:** Developed a vision-language system for real-time object identification, scene understanding and visual reasoning from camera input.
- **SERA:** Developed a Flutter-based AI research assistant integrating Open Router, NVIDIA AI models and REST APIs.
- **Smart Parcel Management System**: Developed a Python-based parcel management system using OpenCV, QR scanning and PyQt.

RESEARCH PUBLICATION

Exploring the Synergy of Automation, Machine Learning, and Embedded Systems: Components, Applications, and Future Trends. International Journal of Progressive Research in Engineering Management and Science (IJPREMS), January 2025.

ACHIEVEMENTS / CERTIFICATIONS

National Level IPR Innovation Challenge 2026 | DataXscape National Level Hackathon Finalist | Young High Flier Award | German A1 Certification | Turtle BOT3 ROS Workshop | 1st place in project expo worth 1000rs |